Market Disparity in the NBA: How does a Team's Salary affect their Wins?

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0.0.1 Introduction

We want to see if an NBA team's market size affects its performance by looking at the win totals of different teams over the past four years, among other variables. To do this, we formed a dataset by merging datasets from multiple websites. The dataset includes the team salaries3 for every team in the past 4 (2021 - 2024) years, their conference2, the population of the city4 they are located in, the net worth of their owner6, what the salary cap was for that year2, a column for salary / salary cap to account for inflation, as well as the win total1 for that season. Our dataset includes a total of 120 different observations (4 seasons, 30 teams). Our outcome variable is wins, and our predictor variables are salary / salary cap, conference, owner net worth, and city population. We want to see if these variables that determine the size of a team's market affect the number of wins a team gets. Big market teams are teams that generally have more wealth and revenue and thus can spend more money, which in turn allows them to sign better players. Big market teams also tend to be in more desirable locations, which serves as another motivator for players to sign there, besides the money. Although the NBA has implemented a salary cap to try to keep a balanced playing field, some teams can afford to spend over the cap limit and pay the penalty, which is a luxury tax. The goal of this report is to find out which predictors are most important when it comes to getting wins and to see if bigger market teams have an advantage when it comes to getting win totals.

0.1 Data Processing

The goal of our project is to look at extremes in the NBA market so we decided to include outliers in our data set that would help indicate and flag teams that spend significantly more or less money as well as teams that have significantly higher or lower win totals.

1 Data Exploration

In order to explore the distribution of the amount of wins, we decided to look at the descriptive statistics of the Wins variable (such as the mean), and we also made a histogram to visualize this data.

Table 1: Descriptive Statistics

Statistic	Value
Minimum	14.00
1st Quarter	32.75
Median	42.00
Mean	39.75
3rd Quarter	48.00

Statistic	Value
Maximum	64.00

And here is the histogram:

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   14.00   32.75   42.00   39.75   48.00   64.00
```

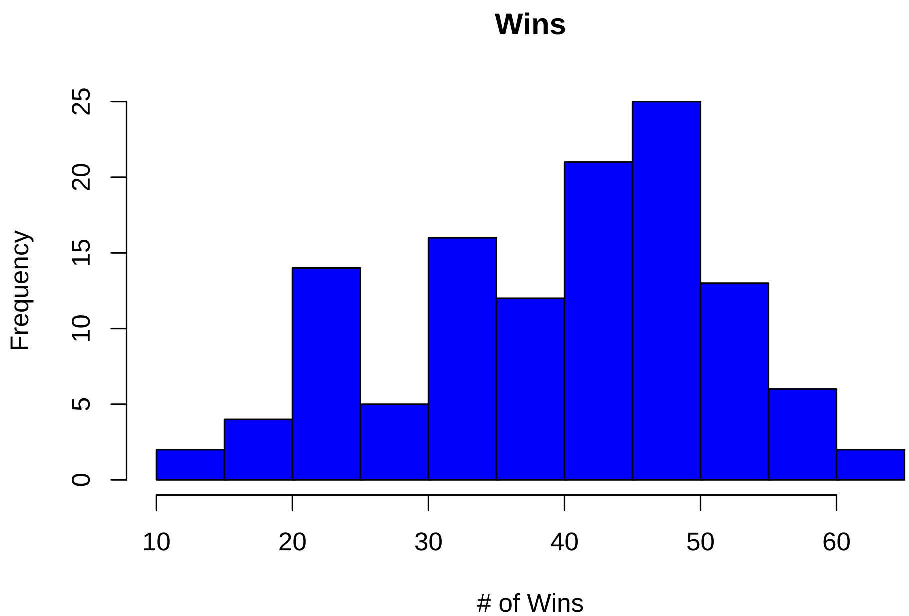


Figure 1: Frequency of Team Wins

From these descriptive statistics and the histogram, we can see that most teams throughout these 4 seasons racked up around 40 wins, with some teams getting much less and more wins with the minimum and maximum values. Now that we saw how the wins were distributed throughout these 4 seasons, we can now create a model that compares a team's number of wins to their salary with respect to the salary cap

1.1 Training and Testing Data Sets

To evaluate the performance of our linear regression model, we split the dataset into training and test sets. We use random sampling to ensure that the data is split without bias. We assign 75% of the data to the training set for model fitting, and the remaining 25% to the test set for model evaluation.

2 Modeling

2.1 Linear Regression Model

In this model, we are investigating the relationship between Salary.Salary.Cap (the outcome variable) and Wins (the predictor). We fit the model using 10-fold cross-validation to evaluate its performance and created a table with the important statistics.

Table 2: Linear Regression Model Summary

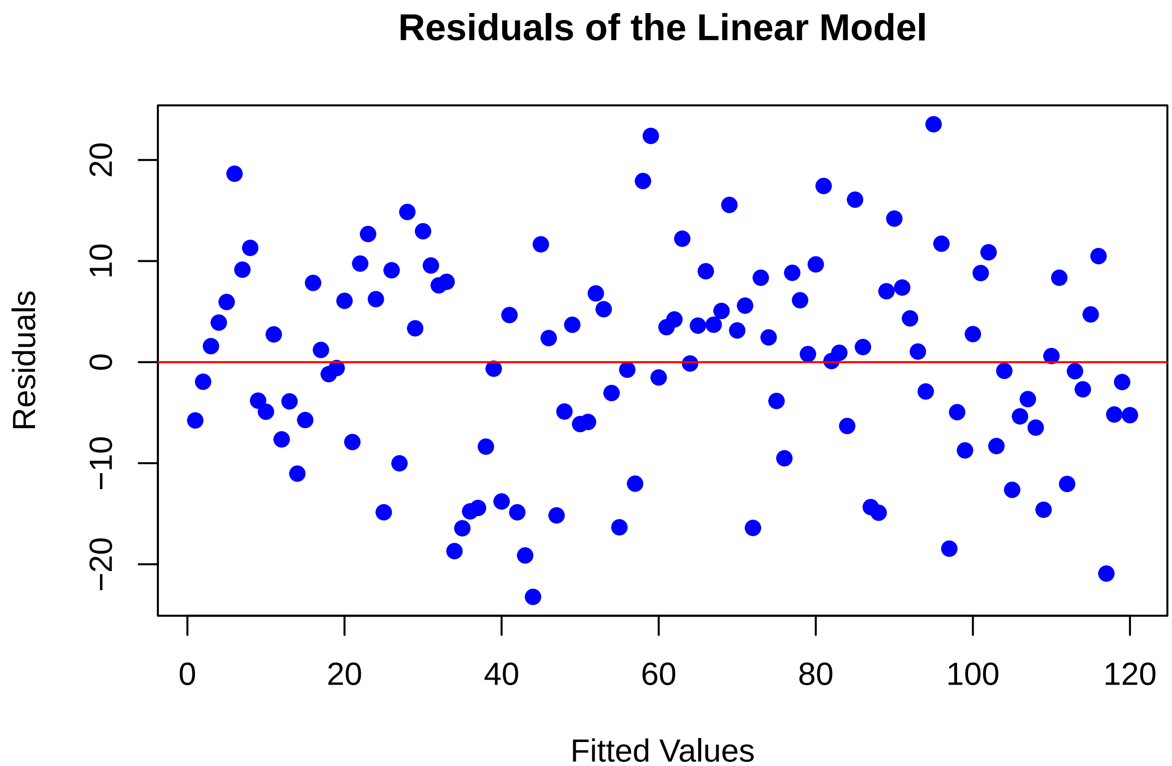
term	estimate	std.error	statistic	p.value
(Intercept)	-3.5232552	7.5793966	-0.4648464	0.6429204
Salary.Salary.Cap	35.4777750	6.3050507	5.6268818	0.0000001
ConferenceWest	0.5681702	1.8919134	0.3003151	0.7644790
Owner.net.worth	0.0000000	0.0000000	-0.6300187	0.5299322
City.Population	0.0000002	0.0000006	0.3603141	0.7192732

As shown in the table above, our model predicts that an increase of one in the Salary / Salary cap variable would increase that team's number of wins by 35. It also predicts that being in the Western Conference increases the number of wins by 1 (rounded). It also shows that the city population and the owner's net worth do not affect the team's wins.

2.1.1 Linear Regression Model Fit Results

#Here is a plot that displays the distribution of the residuals:

```
##
## Call:
## lm(formula = .outcome ~ ., data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -23.2238  -6.1752   0.8625   7.4339  23.5337
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)       -3.523e+00  7.579e+00  -0.465    0.643
## Salary.Salary.Cap  3.548e+01  6.305e+00   5.627 1.31e-07 ***
## ConferenceWest     5.682e-01  1.892e+00   0.300    0.764
## Owner.net.worth   -4.084e-11  6.482e-11  -0.630    0.530
## City.Population    2.184e-07  6.062e-07   0.360    0.719
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.21 on 115 degrees of freedom
## Multiple R-squared:  0.2256, Adjusted R-squared:  0.1987
## F-statistic: 8.375 on 4 and 115 DF,  p-value: 5.767e-06
```



The residuals vs. fitted values plot shows a fairly random scatter around zero, suggesting that the linearity assumption is met. There is no clear pattern in the residuals, which indicates the model fits the data reasonably well. While there may be a slight increase in variance at higher fitted values, it's not severe. Overall, the plot does not reveal major issues with the model.

2.1.2 Linear Regression Model Prediction Performance

For evaluating the model's performance, We used the Root Mean Squared Error (RMSE) and R^2 as performance metrics. We applied the model to the test set to assess how well it generalizes. We also created a plot of predicted vs actual values to show a representation of how well the linear regression model worked.

Table 3: Model Performance on Test Set

RMSE	R_squared
38.39789	0.97975

Use the code chunk below for plots:

```
## `geom_smooth()` using formula = 'y ~ x'
```

The plot and linear regression prove that there is a decently strong linear relationship between the predicted wins and the actual wins, which shows that the model is relatively accurate. From this model, we can see that an increase of one in a team's salary in regards to the salary cap for that season (salary / salary cap) increases their amount of wins by 35, and the p-value for this predictor is less than 0.05 (1.31e-07)so it

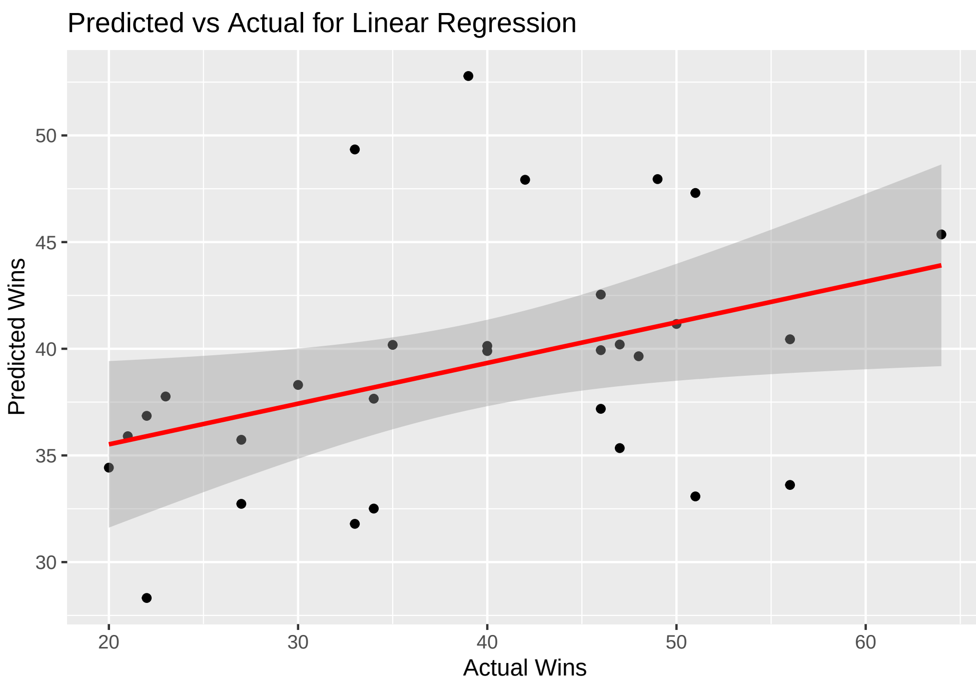


Figure 2: Predicted vs Actual Wins for Linear Regression

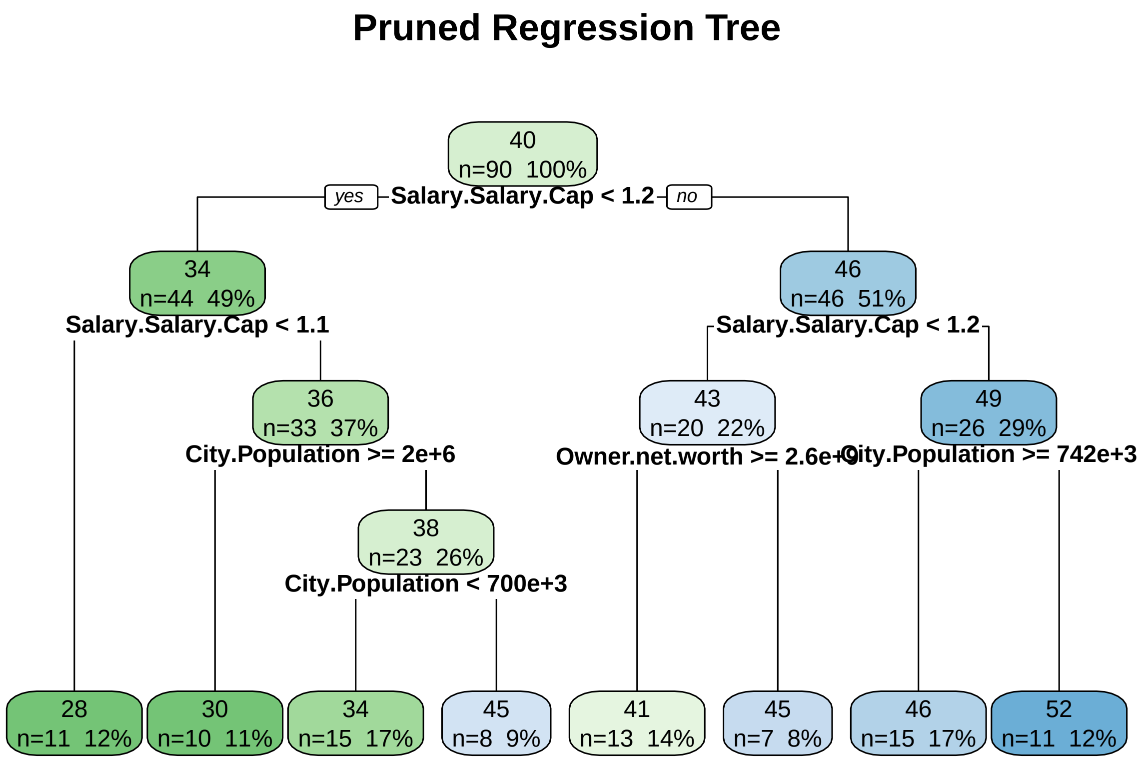
can be concluded that it is statistically significant. This model also shows that the other predictors are not statistically significant, as their p-values are greater than 0.05. So overall, we can see that the most significant variable when it comes to determining a team's number of wins is how large their salary is, as we predicted earlier.

2.2 Regression Tree

Our second model employs a regression tree to predict team wins from the same market-size variables (salary/salary cap, conference, owner net worth, and city population).To better display the data, we pruned the tree based on the best tune for the cp value. This approach provides an interpretable set of decision rules and highlights key breakpoints in how market factors translate into determining each teams win total.

```
## Warning in nominalTrainWorkflow(x = x, y = y, wts = weights, info = trainInfo,
## : There were missing values in resampled performance measures.
```

```
## Warning: cex and tweak both specified, applying both
```



The regression tree makes clear that salary/salary cap is by far the strongest driver of wins, but that its effect is further modulated by market size and owner wealth. At the very top, the tree predicts an average of 40 wins across all 90 training seasons, then immediately splits on the Salary/Salary Cap ratio at 1.2. Teams spending 1.2x the cap fall into the left branch, where they average 34 wins. Teams spending at or above 1.2x the cap occupy the right branch. The tree shows a clear hierarchy in the variables. Salary/salary cap first, then owner wealth and market population. Teams that significantly overspend the cap win roughly 46 games on average, but only the very largest markets or richest owners push that into the low-50s. Low-spending teams almost never reach 30–35 wins.

2.2.1 Model 2 Fit Results

Table 4: Regression Tree CV Results (tuned cp)

cp	RMSE	Rsquared	MAE	RMSESD	RsquaredSD	MAESD
0	9.398	0.448	7.736	4.136	0.289	3.947

Table 5: Variable Importance (Regression Tree)

	Overall
City.Population	100.000
ConferenceWest	0.000
Owner.net.worth	30.666
Salary.Salary.Cap	55.026

2.2.2 Model 2 Prediction Performance

Table 6: Model 2 Performance on Test Set

RMSE	R_squared
39.131	0.549

```
## ‘geom_smooth()‘ using formula = ’y ~ x’
```

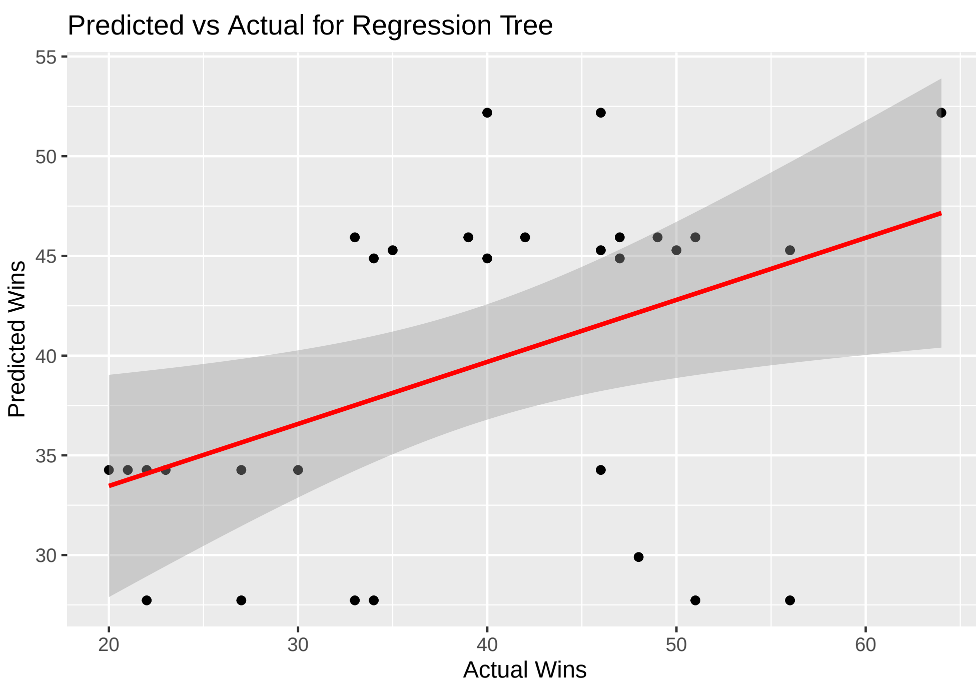


Figure 3: Predicted vs Actual Wins for Regression Tree

3 Result Summary

In this section, you combine the results of both models and compare/contrast their results.

Describe how they fit the training data: did they both consider the same variables “important”? Use a table to summarize this.

Table 7: Variable Importance: Linear vs. Tree

Variable	Linear_Importance	Tree_Importance
Salary.Salary.Cap	100.000	100.000
ConferenceWest	0.000	0.000
Owner.net.worth	6.190	30.666
City.Population	1.126	55.026

Table 8: Test-Set Performance Comparison

Model	RMSE	R_squared
Linear Regression	10.810	0.154
Regression Tree	10.982	0.192

Explain in writing what the table shows.

Describe how each model performed. Use tables and plots. For regression, summarize in a table and/or plot. For classifiers, summarize confusion matrix results in a table, combine both models in an ROC curve.

4 Conclusions

In this section, describe how the results informed your investigation of your research question(s). Did the models help you answer your questions? If not, why do you think that is? Did the results produce any unexpected results or aspects of the data? Did you get any insights from your work that suggest interesting directions to take? How would you go further if you were to continue with this analysis?

You can also mention any difficulties you faced and any threats to the validity o your conclusions.

5 References

#1) Data for wins: https://www.google.com/search?q=nba+23-24+standings&oq=nba+23-24+standi&gs_lcrp=EgZjaHJvbWUqBwgAEAAYgAQyBwgAEAAYgAQyBggBEEUYOTIICAIQABgWGB4yCAgDEAAYFhgeMggIBB
sourceid=chrome&ie=UTF-8#sie=lg;/g/11snv1vp6v;3;/m/05jvx;st;fp;1

#2) Data for Salary Caps: https://www.basketball-reference.com/contracts/salary-cap-history.html #3) Data for Team Salaries and conferences: https://hoopshype.com/salaries/2023-2024/ #4) Data for city populations (most teams): https://worldpopulationreview.com/us-cities #5) Data for Toronto's population: https://www.toronto.ca/city-government/data-research-maps/toronto-at-a-glance/ #6) Data for Owner Net Worths:https://justrichest.com/all-the-nba-team-owners-and-their-net-worth/